

RESEARCH ARTICLE

Application of Traffic Light Control in Oversaturated Urban Network Using Multi-Agent Deep Reinforcement Learning

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ABSTRACT Adaptive traffic signal control techniques have been developed in numerous studies to increase traffic flow efficiency. Using traffic signals to design an adaptive traffic management system is ideal for reducing traffic congestion. Reinforcement learning is a branch of current approaches that try to learn a policy function through a trial-and-error process and maximize the reward through properly adjusted interaction with the learning agent's environment. We propose a traffic signal control architecture for an oversaturated urban network using Deep Q-Network. We have enhanced the learning process by incorporating diverse state information through upstream and downstream detailed traffic states. We conduct experiments on the Simulation of Urban MObility, an open-source traffic simulator that supports large-scale traffic signal control.

INDEX TERMS Multi-agent, oversaturated traffic, reinforcement learning, simulation of urban mobility, traffic signal control.

I. INTRODUCTION

The world's population is growing, and mobility is in high demand. Because of urban space limitations, reducing traffic congestion by road extension is a complex task. On the other hand, congestion results in pollution, unnecessary resource losses, and an increase in the likelihood of traffic accidents. One strategy to deal with the growing number of vehicles that use the current urban networks is to control traffic lights. In this regard, a conventional approach is fixed-time traffic signal control. However, such rigid traffic signal settings would become much more inefficient in a highly dynamic environment.

Adaptive traffic signal control techniques have been developed in numerous studies to increase traffic flow

efficiency. Using traffic signals to design an adaptive traffic management system is ideal for traffic congestion reduction. The most effective approach to improving traffic signal control remains a matter of debate among researchers, largely dependent on the specific traffic conditions in the environment. Up-and-coming options for traffic signal control are to employ learning-based artificial intelligence (AI) algorithms [1], [2], [3], [4], and also using advanced sensor technologies [5], [6]. Reinforcement learning (RL) is a branch of current approaches that tries to learn a policy function through a trial-and-error process and maximize the reward through properly adjusted interaction with the learning agent's environment. When solving the traffic light control problem using reinforcement learning, it is crucial to identify three components: the environment's states, the agent's action space, and the reward for each action [7]. In the past, Q-learning has addressed real-world problems

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but cannot solve high-dimensional issues. One common concern is that the solution space size of a problem increases with each additional feature representing the state. The number of required calculations grows exponentially as the number of state dimension inputs grows. This problem, characterized by Bellman [8] as the curse of dimensionality, has constrained the application of reinforcement learning in many practical domains. The computational complexity of traffic signal control using a traditional reinforcement learning system increases exponentially with extended state space, complicated intersection features, and heterogeneous intersection structures.

Numerous types of research have been conducted by using deep reinforcement learning (DRL) to adapt traffic signal control [9] for a single intersection [10], [11], [12], and multiple intersections [13], [14], [15], [16], [17]. Although reinforcement learning has succeeded in traffic light control in a single intersection, it is still challenging to apply in multiple traffic light controls because the dimensionality of the state and action space grows computationally intractable with the number of intersections. In practical applications, there are many situations where the environment is only partially observable, meaning that agents lack complete information about its state to engage with. This situation can be simulated by using a partially observable Markov decision process (POMDP). Due to multi-agent taxonomies such as the curse of dimensionality and multi-agent credit assignment, deep reinforcement learning for multi-agent environments is inherently more challenging than single-agent environments [18]. The local information about intersections would ultimately lead to modeling complexities as the size of the traffic network grows. Furthermore, the observable region surrounding local intersections impacts traffic control.

Modeling and exploiting other agents' behaviors in a multi-agent system (MAS) has received much attention in the past decade [19]. A multi-agent RL (MARL) algorithm has been proposed as a possible solution to the abovementioned issues. However, the scalability of multi-agent deep reinforcement learning approaches is often limited and vulnerable to partial observability. Centralized learning with decentralized execution is an alternative and popular strategy in which agents are trained simultaneously by using a centralized mechanism via an open communication channel [18], [20]. The centralized control mechanism teaches agents to control the traffic signals across the road network by using global information. On the other hand, using local knowledge, decentralized control techniques model signal light control at multiple intersections as a multi-agent system. Each agent is in charge of a specific intersection and only sees and senses certain aspects of the traffic situation.

In recent times, deep reinforcement learning has gained significant attention, mainly due to the effectiveness of the deep Q-network (DQN) [13], [14], [15], [16]. The Deep Q-learning method [21] modifies the classic Q-learning algorithm with three main innovations: 1) the integration of

a deep convolutional neural network architecture to approximate the Q-function; 2) the utilization of mini-batches comprising randomly chosen training data instead of single-step updates based on the latest experience; and 3) the application of earlier network parameters to estimate Q-values for the next state.

Although many kinds of research have reported multi-agent RL's benefit in urban traffic networks, the situation still needs to be improved when applied to an urban network with oversaturated traffic flow. In this paper, we propose a traffic light control solution for an oversaturated urban road network that considers traffic congestion information of intersections as the states and throughput with backlog penalty in the road network area as the reward for improving the overall performance of the control system.

The main contributions of this paper are summarized as follows:

- 1) Based on multi-agent DQN [21], we propose a traffic signal control architecture for an oversaturated urban network.
- 2) We have enhanced the learning process by incorporating diverse state information through upstream and downstream detailed traffic states.
- 3) We conduct experiments on SUMO [22], an open-source traffic simulator that supports large-scale traffic signal control. We collect spatial-temporal traffic data from the oversaturated urban road network to coordinate multiple agents' collaboration and achieve an improved traffic light control policy.

The remainder of this paper is structured as follows. Section II briefly reviews related works. Section III formulates the problem. Section IV presents a learning architecture in an oversaturated urban network. Section V discusses experimental results based on an urban road network scenario calibrated to represent the area of recurrent traffic gridlock during an evening rush-hour period. Finally, Section VI presents concluding remarks.

II. RELATED WORK

Reinforcement-learning algorithms solve traffic congestion problems, particularly in traffic signal control. Q-learning [23] is a form of reinforcement learning that operates without requiring a model of the environment and that pursues an off-policy approach. On the other hand, the table records all Q-values in the Q-learning process; the rows and columns represent states and actions. A Q-table will be huge when states and actions are numerous. This section will go through the DRL and MARL algorithms that were studied previously.

Based on their objective functions, reinforcement-learning algorithms represent value-based, policy-based, or actor-critic-based approaches. In contrast to conventional RL techniques, DRL methods utilize neural networks to capture learning events, and they include various refinements that are aimed at further improvement. DRL approaches have become popular in urban traffic control research because

nonlinear neural networks can explain real-world traffic. A number of researchers [13], [14], [15], [16] have used the deep Q-learning algorithm for their studies. Such authors have proposed Q-learning by incorporating recurrent neural networks [15] to regulate traffic signals and determine optimal control policies for crucial intersections within road networks. However, reinforcement learning-based systems face challenges that result from the complicated features of intersections, the different structures of various intersections, and the need for dynamic coordination between numerous intersections. For adaptive multi-intersection signal regulation, one study [14] suggested a collaborative deep Q-network to transfer Q-value knowledge, which is denoted as QT-CDQN. Furthermore, value-based approaches are better suited to dealing with discrete state problems like traffic flow than to continuous state problems. Value-based strategies consistently select the action with the highest value and are ineffective when addressing randomness.

Policy-based reinforcement learning has proved effective in dealing with high-dimensional and stochastic continuous action spaces. Such policy-based strategies use an entire state sequence and iteratively update the policy function. Policy-based reinforcement learning establishes the optimal policy through direct modifications to the policy. In contrast, value-based approaches derive the optimal policy implicitly by identifying the optimal value function.

The most recent research on reinforcement learning-based traffic signal control with deep reinforcement learning can be summarized as follows. Each of these approaches depicts a distinct state, reward, or activity. The Q-learning method is challenging to implement in complicated and networked traffic circumstances. A multi-intersection approach has yet to be used in DQN-based studies. When multiple intersections experience congestion, each intersection influences the congestion level of another. Each intersection cannot fully perceive the traffic conditions at all other intersections. Consequently, the situation becomes non-stationary, and individual agents (intersections) can only observe some of the overall traffic conditions. This paper suggests a multi-agent DRL method for determining optimal traffic control rules that maximize throughput in an urban environment. A single agent represents each intersection. The urban road network environment is a multi-agent environment in which one agent's actions influence others' actions, rendering the entire urban road network environment non-stationary from the agents' perspective.

III. SYSTEM MODEL AND FORMULATION

In this section, we first introduce the road network and then formulate state, action, and reward.

A. SIMULATED TRAFFIC NETWORK

This research focuses on a traffic network in a congested urban area. As seen in Fig. 1, the network covers the area of 6 intersections, namely Atthakawi, Klong Toei, Kasemrat, Rama 4, Na Ranong, and Sunlakakon. We have selected

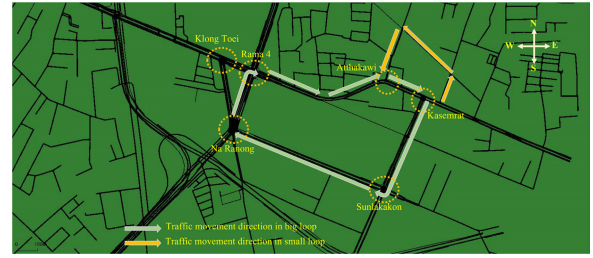


FIGURE 1. A simulated urban road network.

the simulated practical area for two main reasons. Firstly, it is one of the most traffic-congested and has the lowest travel speed in Bangkok, Thailand. Secondly, additional development on-road enlargement is challenging due to the existing infrastructure. Based on the empirical findings from past traffic-collection statistics, the time interval we investigate in our trials is between 3 PM and 7 PM, when traffic congestion is at its worst owing to evening rush hours, and this study aims to improve traffic signal control in this oversaturated urban area. Note that, in Thailand, all road types are left-handed [24], [25].

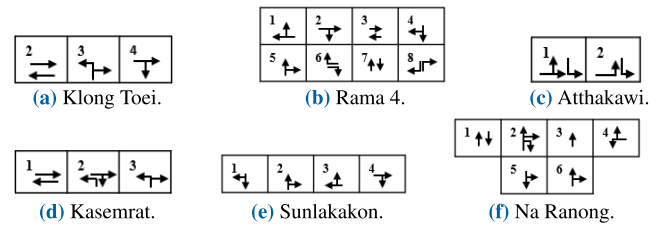


FIGURE 2. Pre-timed traffic signal setting for each intersection in the case study area.

Fig. 2 displays all possible traffic phases with each intersection's pre-timed traffic light control scheme. We have chosen SUMO [22] as a simulator modeling the traffic flow to optimize the fixed-time traffic control strategy and collect state-action pairings for the subsequent training using multi-agent.

B. STATE FORMULATION

The interaction process can be expressed as a Markov decision process (S, A, R, T) [26]. Each intersection is treated as an agent, and each agent is primarily in charge of the local traffic signal located at the corresponding intersection. At discrete time steps, we define the intersection's traffic state s_t , agent chosen action a_t , and the feedback reward r_t ; $t = 1, 2, \dots, T$.

The state is a numerical representation of each intersection agent's observations of the road network environment. If each interconnected intersection in a road network is filled with slow-moving vehicles, then vehicles approaching from opposing directions cannot proceed while getting green traffic light [27]. The region surrounding local intersections has a significant impact on traffic regulation. The duration

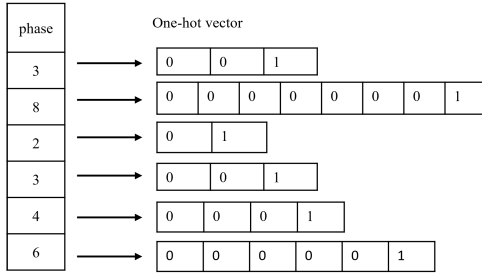


FIGURE 3. Sample encoded one-hot vector for each phase.

of the traffic bottleneck upstream and downstream of the intersection can reflect the congestion situation in space, which defines environmental traffic information. As a result, the observation o of intersection i at time t is defined as

$$o(i,t) = \{j_{1,i,t}^u, \dots, j_{n,i,t}^u, j_{1,i,t}^d, \dots, j_{n,i,t}^d\} \quad (1)$$

where j denotes the length of each traffic jam on upstream with superscript u and downstream with superscript d of each traffic bound of the intersection, and n denotes the number of upstream and downstream queues of the intersection. Our experiment introduces the lane area detectors with 100-m upstream and 100-m downstream on each point of interest region to measure traffic jam lengths. The length of the collected traffic jam represents the traffic condition of every action step. Based on the study [11], we define the action step as an interval of 10 seconds. The observed state as $S_t = \{o_{1,t}, \dots, o_{N,t}\}$ can be determined as the state of the complete road network by all agents. The network consists of N agents. In this method, state S_t reflects the complete environmental information each time t . The different links, such as upstream incoming and downstream outgoing, have varied consequences on each agent's policy. As the multi-agent DQN's input, the state S_t denotes the characteristics of all intersections in the road network.

C. ACTIONS

The agent needs to choose an action to affect the system's dynamics after receiving a representation of the state of the environment $o_t \in S$ at the beginning of the time step t . The action space A collects all possible actions the agents can do. The action of the complete road network $A = \{3, 8, 2, 3, 4, 6\}$ is the set of all traffic signal phases for all agents $i = 1, \dots, 6$, as given in Fig. 2a-f, respectively. For example, Klong Toei, Rama 4, Atthakawi, Kasemrat, Sunlakakon, and Na Ranong intersections use the action that chooses the last-indexed phases 3, 8, 2, 3, 4, and 6 phases, respectively. To represent actions of each intersection i at time t , we use a one-hot vector to encode the value of a_i as shown in Fig. 3. When other phases have been chosen, the one-hot vector values are differently toggled from 0 to 1, with only one 1 per vector at each intersection at a time.

A green traffic light phase indicates that vehicles can proceed through the intersection, while a red signal indicates that vehicles must stop and not proceed through the intersection.

The all-red transient phases are required practically for safety reasons, as they slow and stop traffic for the green phases to take effect. The agents' actions will modify the environment's dynamics. If the selected action a_t is the same as the previous action a_{t-1} , then the traffic light at that time remains unchanged. There is the exception when some downstream corresponding to the previous action is congested (jam length is more than 50 m of measured length 100 m). Here, the phase is forced to transition to a different phase to help ease such downstream congestion. If the chosen action a_t differs from the preceding action a_{t-1} , then the selected action a_t triggers the traffic signal transition. However, this is also the exception when the time since the last phase change is less than the minimum green time duration (15 seconds). Here, the previous phase must remain to prevent phase changes that are too frequent.

Here, we have tried to vary the maximum allowable green time duration t_{max} 1) with the same fixed-timed signal setting, 2) 120 seconds, and 3) 180 seconds. Although the yellow light is practically set to 3 seconds [28], this state has no substantial effect on Bangkok's traffic; thus, the yellow light has been ignored in this study.

D. REWARD FUNCTION

The purpose of traffic signal control in an oversaturated environment is to increase intersection-coordinated throughput and improve the intersection's traffic efficiency. Intuitively, the idea of reward should help evaluate the quality of the agent's actions. The reward report directs agent learning and, in the end, assesses the model's quality. Here, in Eq.(2), f is the number of vehicles on each upcoming edge l that can pass through the intersection m during step t . The total number of vehicles F of all incoming roads, leading to the intersection m , can then be defined as

$$F_{(m,t)} = \sum_{l=1}^E f_{(l,t)} \quad (2)$$

where E represents the number of roads (edges) approaching the intersection m . The higher the reward, the higher the throughput of vehicles that can flow at the intersection. In addition, to quantify effects on the neighbor areas, B , which acts as a penalty, is defined as the number of cars loaded but not departed from the road-entry buffers into the simulated urban network. In the case of multiple intersections, finally, we define the reward as a function of the total throughput of all intersections. The final reward function is defined as

$$R_t = \beta * \sum_{m=1}^M F_{(m,t)} - (1 - \beta) * B \quad (3)$$

where M defines the number of intersections in the environment. In our simulation road network, $M = 6$. Our ultimate goal is to increase throughput by coordinating multiple agents with spatiotemporal relationships. Here, we have introduced a weighting factor β of value between 0 and 1 that determines the relative importance of the terms F and B . When $\beta = 1$,

the term F dominates, and R will be influenced by F of the intersection. Conversely, when $\beta = 0$, the term B dominates, and R will be B , influenced by undeparted vehicles from the road-entry buffers.

IV. LEARNING ARCHITECTURE

A. MULTI-AGENT REINFORCEMENT LEARNING

RL provides the framework through which an agent makes decisions depending on an environmental condition. Agents constantly interact to maximize the objective reward function over time through trial and error. When the state satisfies the Markov property, the interaction process between an agent and the environment is known as a Markov decision process (MDP). In a fully observable MDP, the agent perceives the current state of the environment, represented as s at each time step t . The agent then chooses an action based on a policy, denoted as π , which defines a mapping from states to actions. The environment provides the agent with an immediate reward, represented as r , after which the process iterates until the action that maximizes future rewards has been chosen. The environment transitions to a new state based on the state transition probability function $p(s|a)$, which takes the current state and selected action as inputs.

Estimating the quality of the policy is the main purpose of calculating the Q-function, represented by $Q(s, a)$ in Markov decision processes. In MDPs, the Q-function $Q(s, a)$ can be defined as

$$Q(s, a) = E \left[\sum_{k=0}^{+\infty} \gamma^k r_{t+k} | s_t = s, a_t = a \right] \quad (4)$$

The discount factor, denoted by γ , is a value between 0 and 1 and determines the relative importance of future rewards compared to current rewards. Stating at time t with state $s_t = s$ and action $a_t = a$, the Q function indicates the predicted cumulative discounted reward. To update the Q value for a given condition and action using the Bellman equation, the Q function can be written as

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)] \quad (5)$$

where α denotes the learning rate, s' denotes the next step's state, and a' denotes the next action.

B. DQN-BASED RL

A deep Q-network is a neural network consisting of multiple layers that, when given a specific state s , produces a vector of action values denoted as $Q(s, a; \theta)$, where θ signifies the network's parameters. The DQN algorithm, suggested by [21], relies on two essential components: including a target network and implementing experience replay. The target network mirrors the evaluation network, characterized by parameters θ^- . However, for every τ step, its parameters are updated by copying the corresponding parameters from the evaluation network. This update ensures that $\theta_t^- = \theta_t$ during these updates remains unchanged in all other steps. In the learning process, we perform Q-learning updates on samples

TABLE 1. Configuration settings of road network in SUMO.

Parameter	Value
Intersection size	6
Length of big loop (light green) in Fig. 1	2585 m
Length of small loop (green) in Fig. 1	951 m
Simulation start	54000 s (3 PM)
Simulation end	68400 s (7 PM)
Total simulation time	4 hours
Mobility parameters	same as [24]
Minimum green time duration	15 s
Maximul green time duration	fixed-timed, 120 s, 180 s
Weighting factor (β)	[0.1, 0.3, 0.6, 0.9, 1]

or minibatches of experience (s, a, r, s') , drawn uniformly randomly from the stored sample buffer. The Q-learning update at iteration k involves the utilization of the following loss function:

$$L_k(\theta_k) = E_{(s,a,r,s')} [(r + \gamma \max_{a'} Q(s', a'; \theta_k^-) - Q(s, a; \theta_k))^2] \quad (6)$$

in which γ is the discount factor determining the agent's horizon, θ_k and θ_k^- are the parameters of the evaluation and target Q-networks at iteration k . The parameters of the target network, denoted as θ_k^- , are updated using the Q-network parameters (θ_k) and remain constant between individual updates. The multi-agent DQN algorithm operates with decentralized execution, as illustrated in Fig. 4.

V. EXPERIMENTAL SETTINGS, RESULTS AND DISCUSSION

A. SIMULATION SETTING

The vehicle length is 4.62 meters, and the minimum gap between vehicles is 2.37 meters. The maximum vehicle speed is 12.5 m/s (45km/h). We have used all these mobility parameters that are the same as [24]. For traffic signal timing parameters, the minimum green time is 15 seconds, and the maximum green time has experimented with three settings: 1) the same as the fixed-timed signal setting, 2) 120 seconds, and 3) 180 seconds. Table 1 shows parameters for the road network.

B. TRAFFIC DEMAND SETTING

We have used the traffic demand, only passenger car units (PCUs) surveyed by the Traffic and Transportation Department of Bangkok Metropolitan Administration (BMA) [29].

To investigate oversaturated traffic, we have designed with the following demand settings, also shown in Fig. 5:

- 1) the LOW traffic demand with 12000 vehicles for the simulation network from 3 PM to 7 PM.

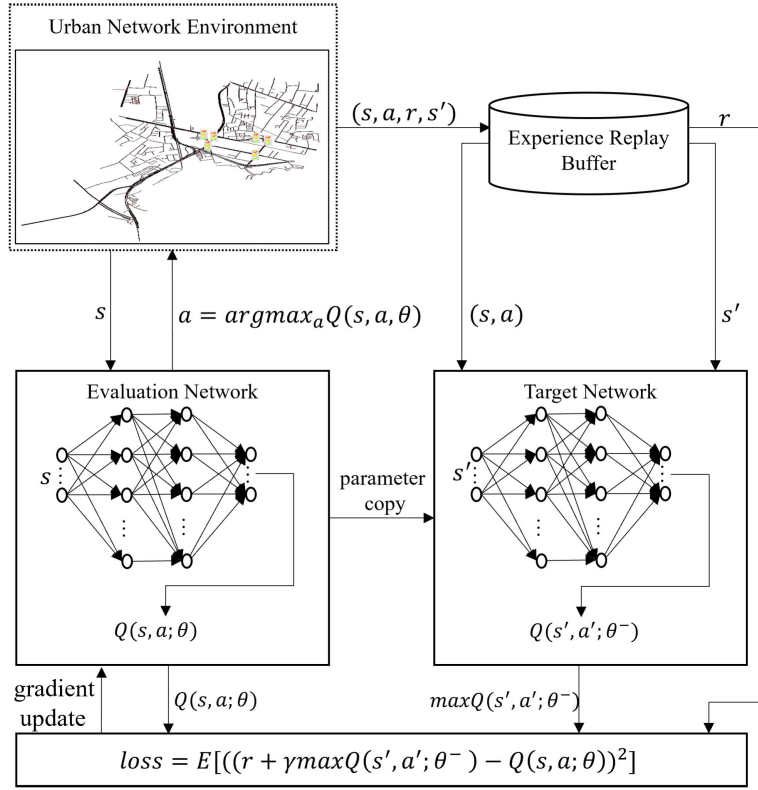


FIGURE 4. DQN-based signal control learning architecture in oversaturated urban network.

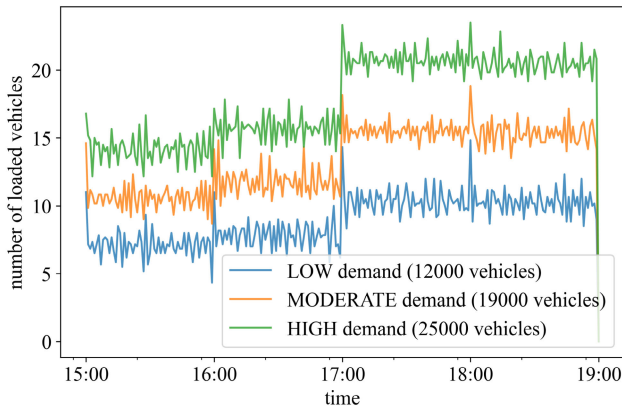


FIGURE 5. Number of loaded vehicles per minute from 3 PM to 7 PM.

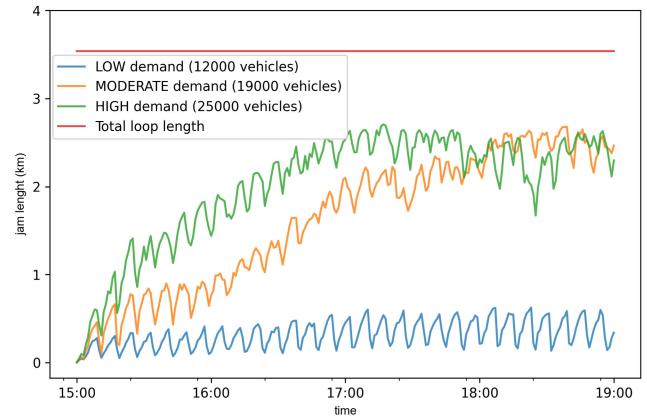


FIGURE 6. Congestion state of case study area based on different number of loaded vehicles from 3 PM to 7 PM in case of fixed-timed signal control.

- 2) the MODERATE traffic demand with 19000 vehicles for the simulation network from 3 PM to 7 PM.
- 3) the HIGH traffic demand with 25000 vehicles for the simulation network from 3 PM to 7 PM.

In earlier studies [27], [30], we have found that bottleneck congestion starts in traffic jams and spreads to upstream and downstream vicinities [31]. Such a circumstance frequently results in gridlock in a complex urban road network, reducing traffic efficiency to a near-halting state. As a result, previous works [27], [30] have proposed gridlock detection and

prediction based on recurrent and non-recurrent congestion using the characteristics of gridlock, which can be assessed by the length of traffic jams and the speed of upstream and downstream links at intersections. According to empirical observations in the current area, as indicated in Fig. 1, daily congestion with bottleneck conditions occurs in the evenings from 3 PM to 7 PM. As seen in Fig. 1, congestion in the small loop impacts the big loop. The roads in the small loop have only one lane in each direction. The bottleneck

TABLE 2. Hyper-parameters for neural network in DQN.

Hyper-parameter	DQN
Epoch length	1440
Episodes	210
Decision epoch	10 s
Optimizer	Adam
Batch size	360 (1 hour)
Discount factor	0.95
Exploration factor	0.9
Learning rate	0.001
No. of hidden layers	2
Neurons in hidden layer	256
Regularization in neural networks	drop out
Replay memory	1500000

occurs in the evening rush hour, which leads to nearly gridlock, and consequently, this also causes the big loop to be congested. Fig. 6 shows the total jam length of both loops in Fig. 1 based on different number of loaded vehicles in the evening peak hours from 3 PM to 7 PM. During the evening rush hours, typically from 3 PM to 7 PM, regarding the congestion spillbacks at the intersection, the length of traffic jams would be relatively longer. If the bottleneck spillbacks at the intersection, the upstream and downstream links of corresponding intersections are congested, and the whole loop is jammed. This overflow in traffic volume contributes significantly to the slowdown and buildup of vehicles on the roadways, creating delays and leading to gridlock in the case study area of Fig. 1.

C. NEURAL NETWORK SETTING

The DQN algorithm has been designed with specific hyperparameters, detailed in Table 2. These hyperparameters include values for learning rates, discount factors, batch sizes, and network architectures. To implement learning architectures in Fig. 4, we have used Python programming language and the PyTorch framework (a popular deep learning library) [32] to build the DQN network. The DQN network has been trained and optimized using the specified hyperparameters. To simulate the multi-agent traffic environment, we have designed an OpenAI Gym environment [33], a widely-used toolkit for developing and comparing reinforcement learning algorithms.

The case study environment allows us to model various traffic scenarios and test the performance of DQN. The scenario setting has been shown in Table. 3. All traffic demand loading has been trained and tested in order in each scenario. For traffic load demands, the first 50 episodes are for training, and the last 20 are for testing. Based on the

practical findings from past traffic collection statistics, the time interval we study in our trials is evening rush hours between 3 PM and 7 PM. Our trials have 1440 steps as epoch length, with a decision epoch interval of 10 s. The computation time of one episode is 12 mins on average using DQN network per each traffic demand loading with each combination of hyper-parameter settings as shown in Table. 3. The computation time reported here is based on the currently used hardware of NVIDIA GeForce GTX 1080 Ti with 11 GB memory. The total step is 302,400 with 210 episodes and 1440 steps within a 4-hour simulation each. To implement DQN, we have referenced from the GitHub repository [34].

D. RESULTS AND DISCUSSIONS

A mix of cooperative and competitive agents coexists within a multi-agent environment. Collaborative agents, or collaborators, aim to align their actions with the controllable agent to achieve shared goals. Conversely, competitive agents, termed opponents, work against the controllable agent's team. Moreover, the agents display a range of offensive and defensive strategies, contributing to the intricate and varied nature of interactions in this multi-agent setting.

The multi-agent DQN enhances the learning framework by controlling the policies in the network, with each agent relying solely on local information. The parameter values are averaged from 1440 steps over a 4-hour simulation period for all results. Fig. 7 compares throughput for all scenarios as shown in Table. 3 with high traffic demand loading. All results in Fig. 7 show varied $\beta = \{0.1, 0.3, 0.6, 0.9, 1\}$ so the learning algorithm can provide high throughput by balancing throughput and backlog as shown in the reward function of section III-D. The results prove that DQN with hyper-parameter settings in scenario 5 with $\beta = 0.9$ provides the highest throughput compared with other hyper-parameter settings. For the following discussion, we chose scenario 5 based on the highest throughput.

Figs. 8-9 shows the performance of multi-agent deep Q-network for selected hyper-parameters setting of scenario 5 with traffic green time duration 180 s in Table. 3. As shown in Figs. 8-9, the deep Q-network significantly enhances network learning performance by enabling agents to independently select actions and receive individual rewards from the environment. We evaluate our proposed model with the parameters throughput and backlog applied in the reward function in section III-D. In addition, QoS (Quality of Service) parameters, such as the mean speed and jam length, have been measured. The multi-agent DQN learning framework has been compared with the baseline fixed-timed traffic signal control currently used by traffic police on road networks.

After finishing the training period starting from 72,000 steps (50 episodes), the multi-agent DQN with $\beta = 0.9$ can provide high throughput (23 PCUs in light traffic demand, 30 PCUs in moderate demand, 35 PCUs in high traffic demand) compared with baseline fixed-timed signal

TABLE 3. The scenario setting with regularization in neural network and maximum green time duration (all dropout regularization with scale 0.5).

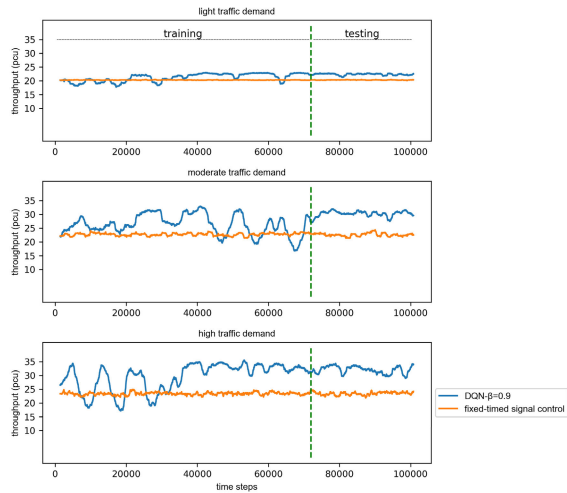
Scenarios	green time duration same as fixed-timed control	green time duration 120 s	green time duration 180 s
1	hidden layers dropout	-	-
2	input and hidden layers dropout	-	-
3	-	hidden layers dropout	-
4	-	input and hidden layers dropout	-
5	-	-	hidden layers dropout
6	-	-	input and hidden layers dropout



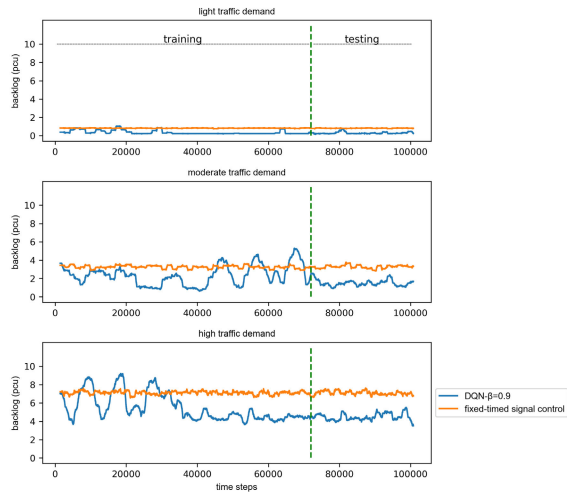
FIGURE 7. Comparison of throughput on testing data with high traffic demand loading by varying β for all scenarios as shown in Table 3.

control. 1 PCU is one vehicle unit that can pass through the intersection. Fig. 8(a) shows the total throughput of all intersections in the case study area. Also, in Fig. 8(b), using a multi-agent deep Q-network control policy with $\beta = 0.9$, there is no backlog in light traffic demand; on average, only 1 PCU and 5 PCUs not departed to enter the road-entry buffer in moderate and high traffic demands, respectively. On the other hand, almost 4 PCUs and 8 PCUs are waiting to enter the road-entry buffer with fixed-timed signal control. The agents can intelligently make adaptive decisions to maximize throughput, and simulation results show that the multi-agent DQN-based framework with a 1% penetration ratio of backlog in reward function with $\beta = 0.9$ achieves better convergence and throughput than baseline fixed-timed signal control.

The QoS parameters total jam length and mean speed of big and small loops, as shown in Fig. 1, have been measured. Using fixed-timed signal control, the mean speed of roads in big and small loops is low, and both loops are high jam as shown in Fig. 9(a). In a multi-agent DQN-based control with a 1% penetration ratio of backlog in the reward function, all vehicles can flow in the loops with a free-flow speed of almost 80 km/h in light and moderate traffic demands and about 50 km/h in high traffic demand as shown in Fig. 9(a). In Fig. 9(b) with high traffic demand loading, loops are very congested with jam length over 2 km (means all vehicles stuck inside the loops) with fixed-timed signal control. The multi-agent DQN-based signal control policy prevents congestion in the loops without jams in low and moderate traffic demands and almost 1 km in high traffic



(a) total throughput for all 6 intersections.



(b) backlog not departed from the road-entry buffer.

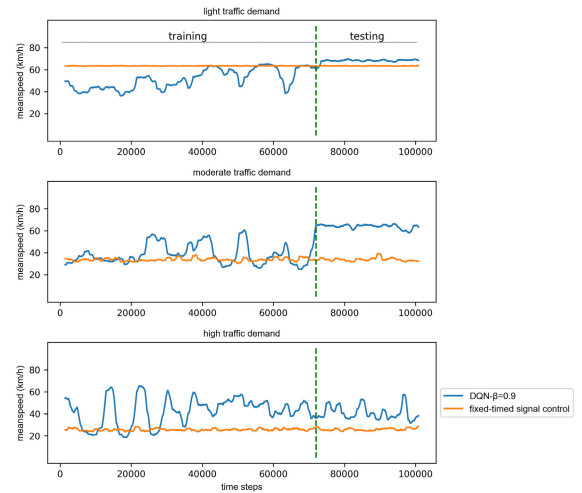
FIGURE 8. Comparison of (a) throughput (b) backlog on training and testing data with different traffic loading.

demands. The agent can acquire an optimal strategy within a high-dimensional state and action space by applying the DQN method.

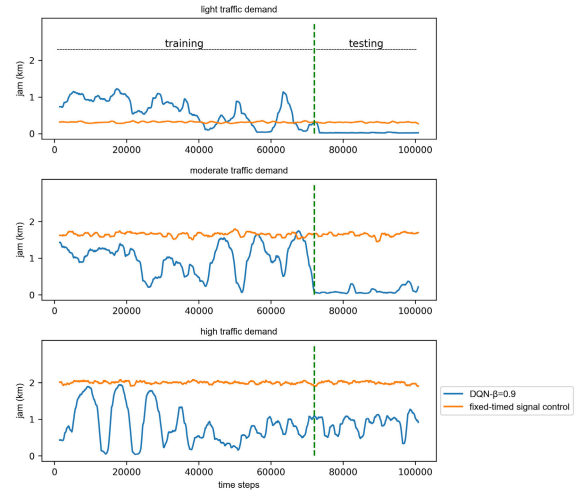
Fig. 10 compares the total throughput of all intersections in the case study area over the jam ratio. The jam ratio has been calculated as the total jam length of both loops over the length of both loops in the case of the study area, as shown in Fig. 1. Vehicles can flow freely inside the loops with light traffic demand using fixed-time signal control and multi-agent DQN strategies. With moderate and high traffic demand loading, from 40% to 60% of the loop length has been occupied by vehicles in fixed-timed signal control. On the other hand, vehicles can still flow easily, and only at most 40% of the loop length has been occupied by vehicles, even in high traffic demand loading using a multi-agent DQN learning policy.

E. PROOF OF CONCEPT

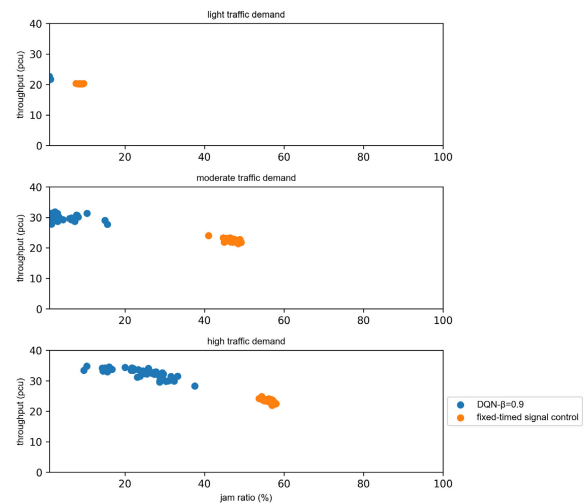
This section presents the proof of concept (POC) of the proposed multi-agent DQN learning policy, which is focused



(a) meanspeed of edges in both loops as shown in Fig. 1.



(b) jam length of edges in both loops as shown in Fig. 1.

FIGURE 9. Comparison of (a) mean speed and (b) total jam length on training and testing data with different traffic loading.**FIGURE 10.** Comparison of throughput over total jam length ratio with different traffic loading.

on determining whether an idea can be turned into a reality. The previous sections have shown the resulting rewards in

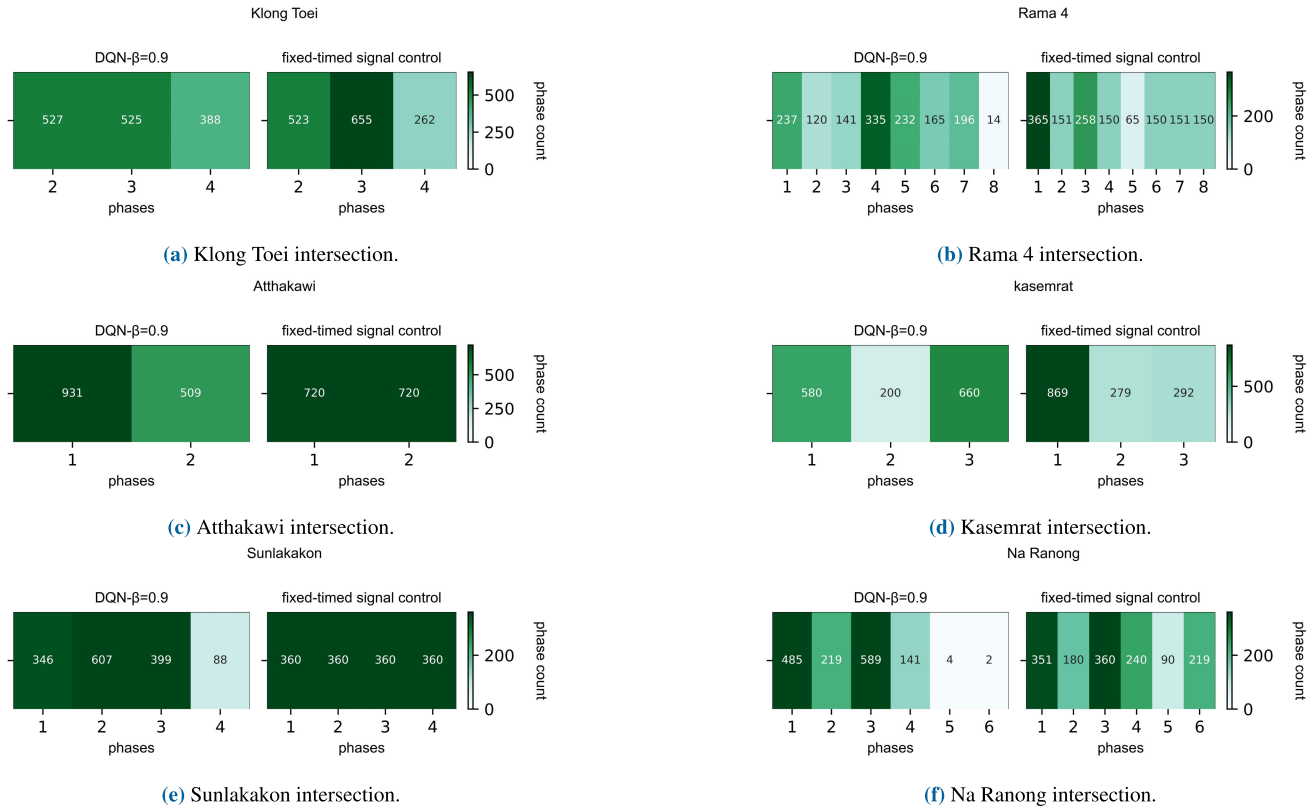


FIGURE 11. Signal control policy comparison of DQN with $\beta = 0.9$ and fixed-timed signal control for light traffic demand.

TABLE 4. The comparison of average total jam and percentage of average total jam over loop length in 4-hour simulation by using fixed-timed signal control and DQN- $\beta = 0.9$.

strategy/metric	fixed-timed signal control			DQN- $\beta = 0.9$		
	LOW demand	MODERATE demand	HIGH demand	LOW demand	MODERATE demand	HIGH demand
total jam length (km)	0.31	1.66	2	0.03	0.11	0.88
percentage of jam/loop length (%)	8.89	46.92	56.68	0.73	3.23	24.91

terms of throughput, backlog, and QoS parameters. Fig. 11 shows the signal control policy comparison of DQN with $\beta = 0.9$ and fixed-timed signal control for light traffic demand by showing the number of phase counts at each intersection in the urban network of the case study area. In multi-agent DQN learning policy, the most used phase sequence is [2, 4, 1, 3, 2, 3] in the order of intersections Klong Toei, Rama 4, Atthakawi, Kasemrat, Sunlakakon, and Na Ranong, as shown in Fig. 12. This phase sequence confirms that the multi-agent DQN learning policy keeps vehicles flowing in the loop by providing high throughput and low traffic jams with high speed.

Table. 4 shows the comparison of average total jam length and percentage of average total jam length over loop length in 4-hour simulation by using fixed-timed signal control and DQN with the weighting factor $\beta = 0.9$. According to the reported results, there is no congestion with a free flow

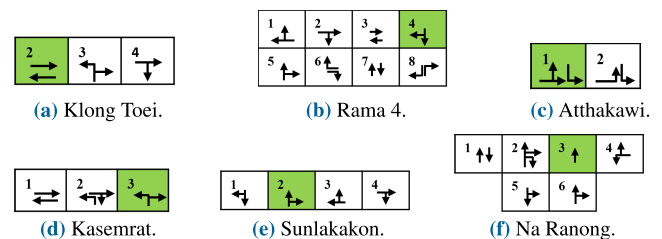


FIGURE 12. Signal control policy of multi-agent DQN with $\beta = 0.9$ for vehicle circulating (mostly used phase number coded with green color).

state in low traffic demand. A traffic jam length of 0.31 km in low traffic demand using fixed-timed signal control has been reduced to 0.03 km, and jam over loop length of 8.89% has decreased to 0.73% using DQN. Likewise, the jam length from 1.66 km in moderate traffic demand has

been reduced to 0.11 km, and over 40% of the jam length can be reduced with the help of the DQN learning strategy. In high traffic demand using fixed-timed signal control, the total jam length is 2 km, and it is about 56.68% of the loop has been occupied by vehicles. On the other hand, with the DQN learning strategy, 0.88 km total jam length and only 24.91% of the loop can be congested. Using the signal control policy of DQN as shown in Fig. 12, agents employ control policies to address traffic congestion by dynamically adjusting traffic flow and signal timings to optimize overall system performance and minimize delays. Through iterative learning and adaptation, these policies enable agents to respond effectively to changing traffic conditions, thereby improving efficiency and alleviating congestion on road networks.

VI. CONCLUSION

Deep Q-Network is a groundbreaking algorithm that combines deep neural networks with Q-learning for reinforcement learning tasks. Its ability to learn optimal policies in complex environments has made it a widely used algorithm in traffic signal control. DQN can generalize learned policies to unseen states by varying traffic demand loading, allowing for better adaptation and decision-making in our tested scenarios. This study has applied spatial-temporal traffic data from the oversaturated urban road network to coordinate multiple agents' collaboration and achieve an improved traffic light control policy. The proposed multi-agent DQN signal control policy can provide high throughput in an oversaturated urban network. This study suggests a multi-agent DQN for determining optimal traffic control rules that maximize throughput with low traffic jams in an urban environment.

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